

# 1. Log Correlation

Log correlation is the process of analyzing and linking events from different log sources to obtain a complete understanding of system activities and security incidents. Since a single event may generate records across multiple devices and applications, correlating logs helps security analysts identify relationships between these events and reconstruct what actually occurred.

Logs can be collected from sources such as firewalls, web servers, operating systems, intrusion detection systems, and authentication servers. By comparing timestamps, IP addresses, usernames, hostnames, and event types, analysts can connect related activities and create a unified view of an incident.

For example, a firewall log may show a connection from a specific IP address, a web server log may record requests from the same IP address, and an authentication log may show login attempts from that source. When these events are correlated, they provide a clearer picture of user behavior or malicious activity than any single log source could provide independently.

The main benefits of log correlation include:

- Improved visibility of security events.
  - Faster incident detection and investigation.
  - Identification of attack patterns and suspicious activities.
  - Creation of accurate timelines for forensic analysis.
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## 2. Identifying an Attack Chain from Logs

An attack chain refers to the sequence of actions performed by an attacker during a cyberattack. Security logs provide evidence of these actions, allowing analysts to identify how the attack progressed from initial access to its final objective.

To identify an attack chain, analysts examine logs for indicators of different attack stages. Common stages include:

1. **Reconnaissance** – The attacker gathers information about the target through activities such as port scanning or vulnerability scanning.
2. **Initial Access** – The attacker gains entry into the system by exploiting vulnerabilities or using stolen credentials.

3. **Execution** – Malicious commands or programs are executed on the compromised system.
4. **Persistence** – Mechanisms are established to maintain access, such as creating new user accounts or scheduled tasks.
5. **Privilege Escalation** – The attacker attempts to gain higher-level permissions.
6. **Lateral Movement** – The attacker moves across systems within the network.
7. **Data Collection and Exfiltration** – Sensitive information is gathered and transferred outside the organization.

By examining logs and arranging events in chronological order, analysts can determine which attack stages occurred and how the attacker moved through the environment.

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### 3. Full Attack Sequence Reconstruction

Attack sequence reconstruction is the process of creating a detailed timeline of an incident based on correlated logs and identified attack stages. The objective is to clearly describe the actions performed by the attacker from beginning to end.

#### Example Attack Sequence

1. At 09:00 AM, multiple port scan attempts were detected from an external IP address, indicating reconnaissance activity.
2. At 09:05 AM, the attacker launched several SQL injection attempts against the organization's web application.
3. At 09:06 AM, one of the SQL injection attacks succeeded, allowing the attacker to upload a web shell to the server.
4. At 09:10 AM, commands were executed through the web shell, providing remote access to the compromised system.
5. At 09:15 AM, a new administrator account was created to maintain persistent access.
6. At 09:20 AM, the attacker accessed and collected confidential files stored on the server.
7. At 09:30 AM, a large volume of data was transmitted to an external destination, indicating data exfiltration.
8. Following the data transfer, the attacker retained administrative access for potential future use.

#### Timeline of Events

| Time | Activity |
|------|----------|
|------|----------|

09:00 AM Port scanning detected

09:05 AM SQL injection attempts

09:06 AM Successful exploitation and web shell  
upload

09:10 AM Remote command execution

09:15 AM Administrator account creation

09:20 AM Access to sensitive files

09:30 AM Data exfiltration

In conclusion, log correlation enables analysts to connect related events from multiple sources, attack chain identification helps determine the progression of an attack, and attack sequence reconstruction provides a complete chronological account of the incident. Together, these techniques play a crucial role in cybersecurity investigations, incident response, and digital forensics.